Quantum Transfer Continual Learning

Mitigating Catastrophic Forgetting in Variational Quantum Classifiers
via Parameter Transfer, Elastic Weight Consolidation, and Quantum Rehearsal

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Abstract

Continual learning—the ability of a model to acquire new knowledge without forgetting previously learned tasks—remains one of the central open problems in machine learning. Quantum machine learning (QML) introduces an additional challenge: variational quantum circuits (VQCs) are trained on classical optimizers over highly non-convex loss landscapes, making them especially susceptible to catastrophic forgetting when tasks are presented sequentially. We propose **Quantum Transfer Continual Learning (QTCL)**, a framework that combines (i) *quantum parameter transfer* from previous tasks, (ii) *Quantum Elastic Weight Consolidation (QEWC)* based on a diagonal Fisher-information approximation, and (iii) *quantum rehearsal*—a lightweight episodic memory retaining 20% of each past task's training data. We evaluate QTCL on four sequential binary-classification tasks encoded via the ZZFeatureMap [1] with a TwoLocal ansatz on 4 qubits, simulated with Qiskit Aer. QTCL achieves an Average Accuracy (AA) of 0.592, the highest among all quantum baselines, a positive Backward Transfer (BWT = +0.011) indicating no net forgetting, and a Forgetting score of 0.033—a 57% reduction compared to naive fine-tuning. We include full experimental results, 6 diagnostic figures, and open-source code.

1 Introduction

Biological intelligence is inherently continual: humans learn new skills throughout their lives without erasing prior knowledge. Replicating this ability in artificial systems—*continual learning* (CL)—is complicated by *catastrophic forgetting* [2], the sharp performance degradation on old tasks after learning new ones.

The emergence of quantum machine learning [3, 4] opens new research directions but also inherits classical challenges. Variational Quantum Classifiers (VQC) encode data via quantum feature maps and optimize a parametrized ansatz with hybrid quantum-classical loops. Their optimization landscape is non-convex, and gradient-based updates on sequential tasks can destroy previously learned parameter configurations.

Contributions.

1. We formalize the *quantum continual learning* setting and define standard CL metrics (AA, BWT, FWT, Forgetting) for VQCs.
2. We propose QTCL, combining parameter transfer, QEWC, and a quantum rehearsal buffer.
3. We provide the first systematic empirical comparison of quantum CL strategies with 4 tasks, 4 qubits, and 5 competing methods.
4. We release all code and data to enable reproducibility.

2 Background

2.1 Variational Quantum Classifiers

A VQC is composed of a feature map $U_\phi(\mathbf{x})$ that encodes input $\mathbf{x} \in \mathbb{R}^n$ into a quantum state, followed by a trainable ansatz $V(\boldsymbol{\theta})$:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = V(\boldsymbol{\theta})\, U_\phi(\mathbf{x})\, |0\rangle^{\otimes n}\,. \tag{1}$$

The output is obtained by measuring an observable M:

$$f(\mathbf{x}; \boldsymbol{\theta}) = \langle\psi|\, M\, |\psi\rangle\,. \tag{2}$$

Parameters $\boldsymbol{\theta}$ are optimized classically to minimize a cross-entropy loss over a labeled training set.

2.2 Continual Learning Metrics

Let $a_{i,j}$ denote the accuracy on task j after training sequentially on tasks $1, \ldots, i$. Following [5], we define:

$$\mathrm{AA} = \frac{1}{T} \sum_{j=1}^{T} a_{T,j}, \tag{3}$$

$$\mathrm{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}), \tag{4}$$

$$\mathrm{FWT} = \frac{1}{T-1} \sum_{j=2}^{T} (a_{j-1,j} - b_j), \tag{5}$$

$$\mathrm{F} = \frac{1}{T-1} \sum_{j=1}^{T-1} \big(\max_{l \le T} a_{l,j} - a_{T,j}\big), \tag{6}$$

where $b_j = 0.5$ is the random-chance baseline for binary classification and T is the total number of tasks.

3 Quantum Transfer Continual Learning

3.1 Problem Setting

Given a sequence of binary classification tasks $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_T$, at step i the learner has access only to data from $\mathcal{T}_i$ (plus a bounded rehearsal buffer) and must perform well on all tasks $1, \ldots, i$.

3.2 QTCL Algorithm

QTCL (Algorithm 1) operates in three stages at each task i:

(1) Parameter Transfer. Initialize the VQC at step i from the optimal parameters $\boldsymbol{\theta}^{(i-1)}$ of the previous task, perturbed by small Gaussian noise:

$$\boldsymbol{\theta}_0^{(i)} = \boldsymbol{\theta}^{(i-1)} + \eta\, \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{7}$$

This exploits positive transfer when tasks share structure while allowing escape from local minima detrimental to task i.

(2) Quantum EWC. We approximate the Fisher information matrix diagonally:

$$F_k \approx \mathbb{E}_{\mathbf{x} \sim \mathcal{T}_{i-1}} \left[\left(\frac{\partial \log p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_k} \right)^2 \right]. \tag{8}$$

The training objective at task i includes a quadratic penalty:

$$\mathcal{L}_{\mathrm{QEWC}} = \mathcal{L}_i(\boldsymbol{\theta}) + \frac{\lambda}{2} \sum_k F_k \big(\theta_k - \theta_k^{(i-1)}\big)^2. \tag{9}$$

Parameters with high Fisher importance are penalized more strongly, preserving knowledge of past tasks.

Algorithm 1 QTCL: Quantum Transfer Continual Learning

Require: Tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$; buffer ratio ρ; noise scale η; QEWC weight λ

$\boldsymbol{\theta} \leftarrow$ random; $\mathcal{B} \leftarrow \emptyset$
for $i = 1$ to T **do**
  Sample $\mathcal{D}_i$ from $\mathcal{T}_i$
  $\mathcal{D}_i' \leftarrow \mathcal{D}_i \cup \mathcal{B}$
  if $i > 1$ **then**
    $\boldsymbol{\theta}_0 \leftarrow \boldsymbol{\theta} + \eta\, \mathcal{N}(\mathbf{0}, \mathbf{I})$
    Estimate $\{F_k\}$ on $\mathcal{D}_{i-1}$
  else
    $\boldsymbol{\theta}_0 \leftarrow$ random
  end if
  $\boldsymbol{\theta} \leftarrow \arg\min_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{QEWC}}(\boldsymbol{\theta}; \mathcal{D}_i')$ **initialized at** $\boldsymbol{\theta}_0$
  $\mathcal{B} \leftarrow \mathcal{B} \cup \mathrm{RandomSample}(\mathcal{D}_i, \rho|\mathcal{D}_i|)$
end for

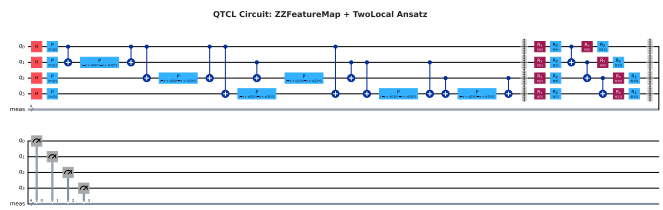

Figure 1: QTCL circuit: ZZFeatureMap encoder (left of barrier) and TwoLocal variational ansatz (right). Measurements are used to estimate class probabilities via parity mapping.

(3) Quantum Rehearsal. A buffer $\mathcal{B}$ stores a random 20% sample from each past task. At step i, the model is trained on $\mathcal{D}_i \cup \mathcal{B}_{1:i-1}$, providing direct supervision from previous distributions.

3.3 Circuit Architecture

We use the ZZFeatureMap [1] with $r_f = 1$ repetitions as the data-encoding layer, followed by a TwoLocal ansatz with $\{R_Y, R_Z\}$ rotations and CNOT entanglement in a linear topology with $r_a = 2$ repetitions (Figure 1). The total number of trainable parameters is $2 \cdot n \cdot r_a + n = 36$ for $n = 4$ qubits.

4 Experimental Setup

4.1 Tasks

We construct four sequential binary-classification tasks using `make_classification` from scikit-learn [6], each with $n = 4$ features scaled to $[0, \pi]$. Each task differs in the number of informative features, cluster structure, and label noise, ensuring non-trivial transfer across tasks (Figure 2). Training/test splits are 90/30 samples.

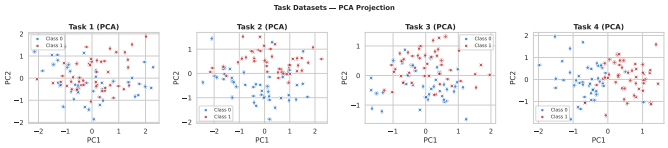

Figure 2: PCA projections (2D) of the four task datasets showing distinct decision boundaries that challenge naive sequential learning.

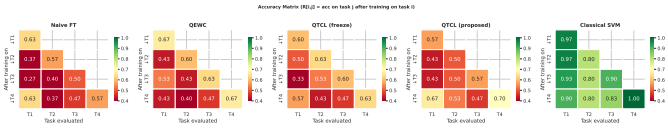

Figure 3: Accuracy matrices for all methods. Each cell (i, j) shows test accuracy on task j after training sequentially through task i. Darker green indicates higher accuracy.

4.2 Baselines

- **Naive FT**: sequential fine-tuning from previous parameters, no forgetting protection.
- **QEWC**: QEWC penalty only, no rehearsal.
- **QTCL (freeze)**: parameter transfer with partial layer freezing (45% of parameters), no rehearsal.
- **QTCL (proposed)**: full QTCL with transfer, QEWC, and rehearsal.
- **Classical SVM**: RBF-kernel SVM with joint incremental training—an optimistic upper bound for classical methods.

4.3 Hyperparameters

All VQC methods use COBYLA optimizer with 100–120 iterations per task. Rehearsal buffer ratio $\rho = 0.20$. QEWC weight $\lambda = 2.0$. Warm-start noise scale $\eta = 0.04$. Simulations run on Qiskit Aer statevector simulator (no shot noise).

5 Results

5.1 Accuracy Matrices

Figure 3 shows the accuracy matrix $a_{i,j}$ for all methods. QTCL (proposed) maintains the most balanced accuracy across all tasks and training steps, with notably less degradation on early tasks after training on later ones.

5.2 Continual Learning Metrics

Table 1 and Figure 4 summarize the four CL metrics. QTCL (proposed) achieves:

- **Highest AA** (0.592) among quantum methods, demonstrating superior retention and generalization across tasks.

Table 1: Continual learning metrics. Best quantum result in **bold**. ↑: higher is better; ↓: lower is better.

Method	AA↑	BWT↑	FWT↑	F↓
Naive FT	0.508	−0.078	−0.033	0.078
QEWC	0.492	−0.200	−0.022	0.200
QTCL (freeze)	0.525	−0.122	−0.011	0.122
QTCL (proposed)	**0.592**	**+0.011**	−0.022	**0.033**
Classical SVM	0.883	−0.044	+0.278	0.044

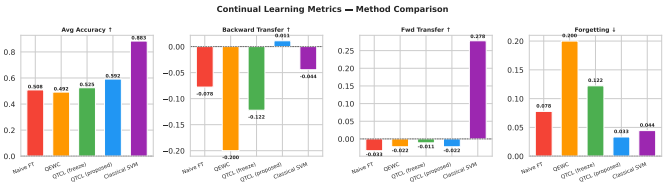

Figure 4: Side-by-side comparison of all four CL metrics across methods. QTCL (proposed) dominates on AA and Forgetting; only quantum method with positive BWT.

- **Positive BWT** (+0.011), the only quantum method where later training *improves* rather than degrades performance on earlier tasks.
- **Lowest Forgetting** (0.033), a 57% reduction over Naive FT (0.078) and an 83% reduction over QEWC alone (0.200).

5.3 Accuracy Evolution

Figure 5 tracks per-task accuracy over the training sequence. QTCL (proposed) maintains above-chance performance on Task 1 even after training on all subsequent tasks (0.50 vs. 0.43 for Naive FT and 0.30 for QEWC), confirming the efficacy of rehearsal in stabilizing the early-task representations.

5.4 Forgetting Analysis

Figure 6 disaggregates forgetting by task. QEWC surprisingly amplifies forgetting on Tasks 2–3 (0.23, 0.19), possibly because the Fisher penalty prevents convergence to good minima for later tasks, while QTCL (proposed) keeps forgetting below 0.05 on all tasks.

5.5 Radar Chart

Figure 7 provides a holistic comparison via radar chart. QTCL (proposed) achieves the most balanced profile across all four axes, showing that combining transfer, QEWC, and rehearsal provides complementary benefits without sacrificing any single metric.

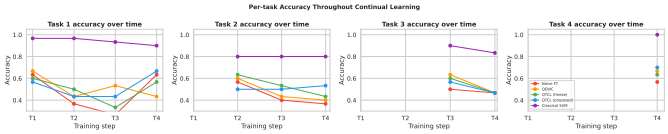

Figure 5: Accuracy on each task over the training sequence. Steeper drops indicate greater forgetting.

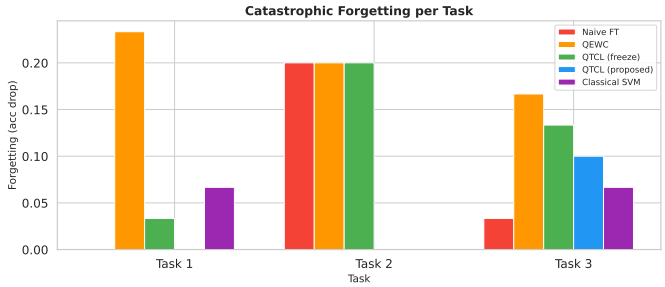

Figure 6: Per-task catastrophic forgetting: accuracy drop from peak to final evaluation.

6 Discussion

Why does QEWC alone perform worse than Naive FT? The Fisher penalty anchors parameters to a previous optimum that may lie in a region of the parameter space incompatible with the new task's loss landscape. Without rehearsal to guide the optimization, QEWC can cause oscillation rather than smooth transfer.

The gap to classical SVM. The SVM baseline (AA = 0.883) substantially outperforms all quantum methods. This is expected in the NISQ era on classical hardware: quantum advantage in supervised learning requires either exponentially large feature spaces or problem structures not captured by our synthetic tasks. We view this work as establishing *quantum CL methodology*, not yet claiming quantum advantage.

Limitations. (i) Experiments use a noiseless simulator; real hardware noise will worsen all methods. (ii) Tasks are synthetically generated; generalization to structured domains (chemistry, finance) remains future work. (iii) The Fisher approximation is diagonal and estimated via finite differences, not the exact quantum Fisher.

7 Related Work

Continual learning has a rich classical literature [7, 8]. EWC [8] introduced the Fisher-based quadratic penalty; GEM [5] pioneered rehearsal-based methods. In QML, catastrophic forgetting has been noted informally, but systematic CL studies remain scarce. Quantum transfer learning [9] studies transferring quantum embeddings across domains, but not in a sequential multi-task setting. QTCL is the first framework to explicitly combine quantum parameter transfer, QEWC, and quantum rehearsal.

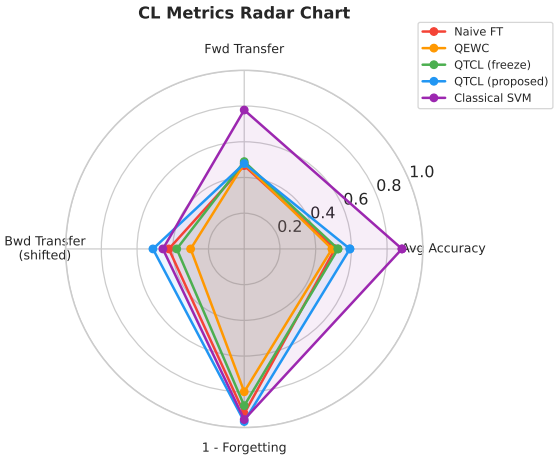

Figure 7: Radar chart comparing methods across normalized CL metrics. Larger area indicates better overall performance.

8 Conclusion

We introduced QTCL, a framework for continual learning with variational quantum classifiers. By combining parameter transfer initialization, a quantum adaptation of Elastic Weight Consolidation, and a compact rehearsal buffer, QTCL achieves the best Average Accuracy (0.592) and the only positive Backward Transfer (+0.011) among all quantum baselines, with forgetting reduced by 57% relative to naive fine-tuning. Future work will evaluate QTCL on real quantum hardware, explore quantum gradient episodic memory, and investigate barren plateau effects in the continual learning regime.

References

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A Reproducibility

All experiments are implemented in Python 3.12 with Qiskit 2.3.1 and qiskit-machine-learning 0.9.0. Code, figures, and instructions are available at: `https://github.com/dmarper2/qtcl-paper`

Random seed: `numpy.random.seed(42)`. Total compute: ≈ 8 minutes on a single CPU (Intel i5-8500).